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Instruction

What the question is

NEW LAST MIN CHANGES

Turing Machine State Diagrams from TM Variants

Sketch a state diagram of a TM to recognize each of the languages described in: 3.8b (Assuming a leading blank)

3.8 Give implementation-level descriptions of Turing machines that decide the following languages over the alphabet $\{0,1\}$.

[redacted]

b. $\{w \mid w \text{ contains twice as many 0s as 1s}\}$

[redacted]

Sketch a state diagram of a TM to recognize each of the languages described in: 3.8b and 3.8c (Assuming a leading blank)

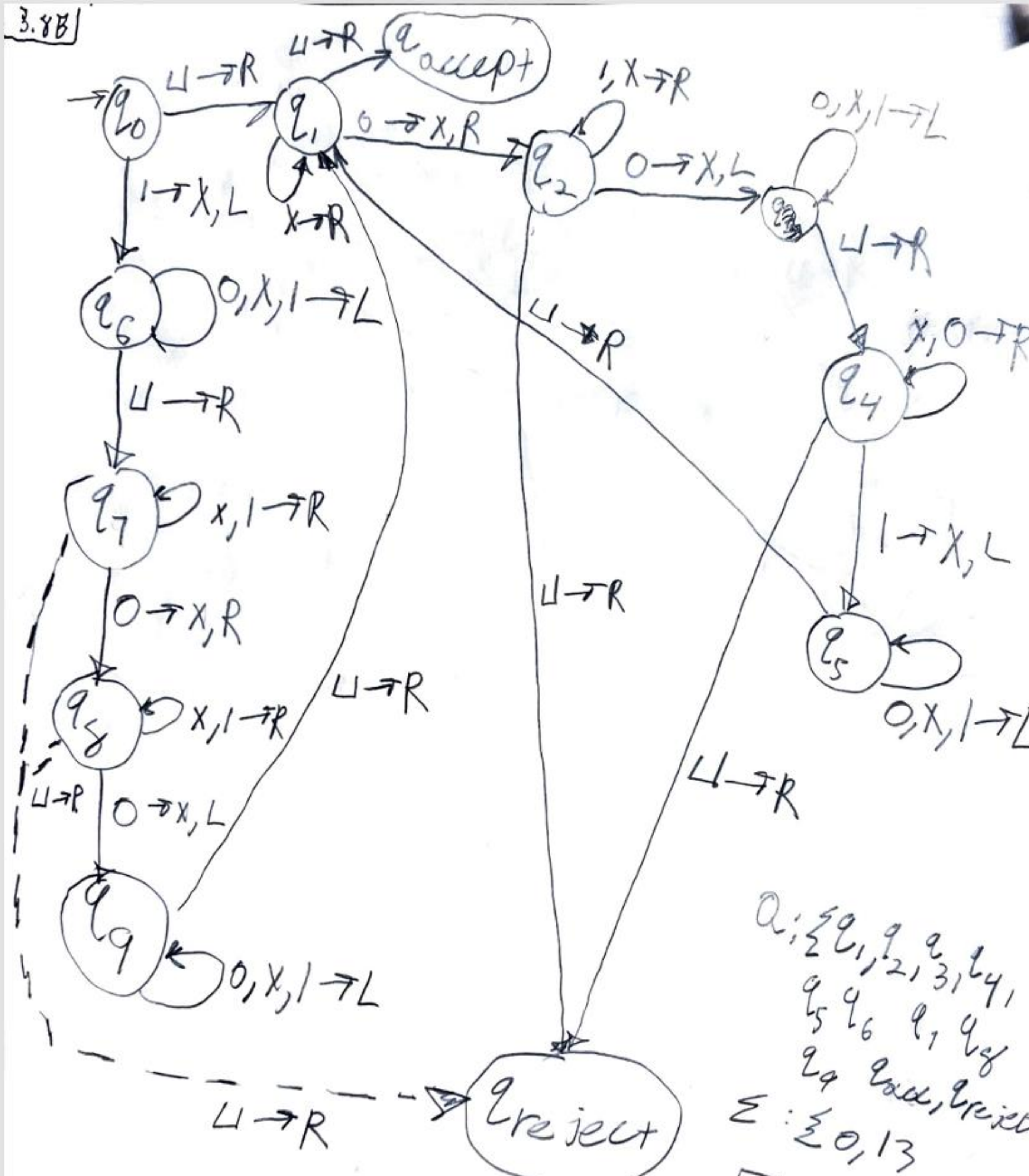
3.8 Give implementation-level descriptions of Turing machines that decide the following languages over the alphabet $\{0,1\}$.

[redacted]

b. $\{w \mid w \text{ contains twice as many 0s as 1s}\}$ Prev. done

c. $\{w \mid w \text{ does not contain twice as many 0s as 1s}\}$

3.8B



Q: $\{q_1, q_2, q_3, q_4, q_5\ q_6\ q_7, q_8\ q_9\ q_{acc}, q_{reject}\}$

Σ : $\{0, 1\}$

Γ : $\{0, 1, X, ⊔\}$

q_0 : q_0

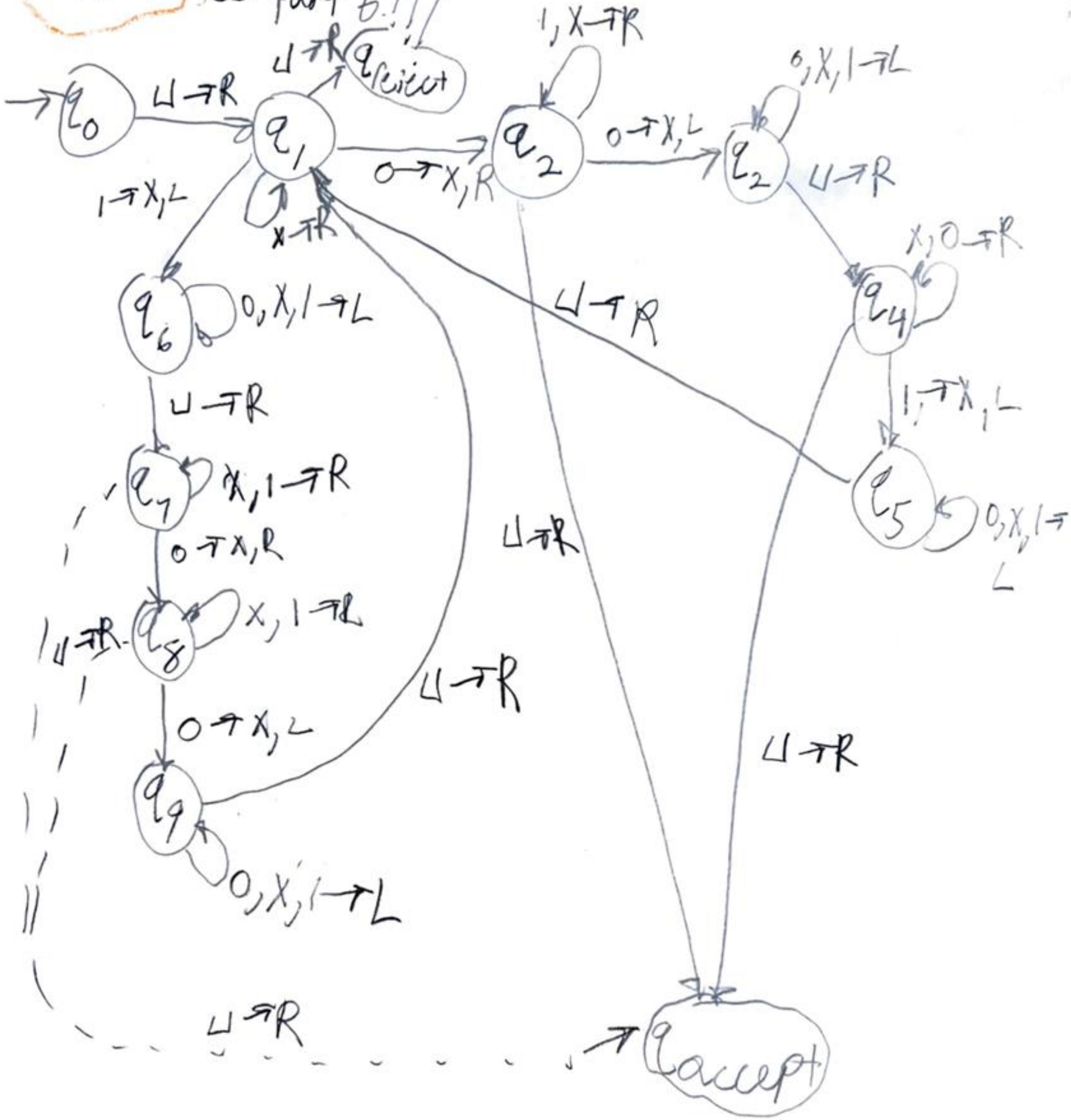
q_{accept} : q_{accets}

$\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$

q_{reject} : q_{reject}

credit to Office hours since I didn't know where to start.

3.8c) For part c you just got to flip accept & reject see part b!!!



see 3.8b
for documentati

What I did previously:

3.8b) Given Lang $L_2 = \{w \mid w$ contains 2x as many 0's as 1's over the ~~the~~ alphabet $\{0,1\}$

Let M_2 be the machine that decides the Lang L. The implementation lvl description of M_2 is as follows:

M_2 = On input w:

St 1:
S Scan the tape for the 1st unmarked 1, mark it. If there are no unmarked 1's, go to Step 5. else Place the head at the Start of the tape.

Step 2:
Scan the unmarked 0 is found in the tape & mark it. If there are no 0's, reject.

Step 3: Scan the tape again till the unmarked 0's is said to be found & mark them. If there is no unmarke 0's → reject.

Step 4: Move head back to the front of the tap & repeat St 1.

5 Place the head back to the front of the tape to se if there are any unmark 0's are found. If none are found, then accept. Else → reject.

TM Variant: 3.11

3.11 A ***Turing machine with doubly infinite tape*** is similar to an ordinary Turing machine, but its tape is infinite to the left as well as to the right. The tape is initially filled with blanks except for the portion that contains the input. Computation is defined as usual except that the head never encounters an end to the tape as it moves leftward. Show that this type of Turing machine recognizes the class of Turing-recognizable languages.

3.11 To prove the double infinite tape T.M D = the ... T.M M. To show equivalence between D & M. we need to show 2 things

- Any Lang L that can be recognized by D can be recognized by M
- Any Lang L that can be recognized by M can be recognized by D as well.

1) Sim M by D

1) Mark left hand end of D
2) Prevent D from moving it's head to the left of mark

D sim M.

2) Sim D → M:

To sim the double ∞ tape turing machine D by an ordinary TM M, sim it w/ a 2 tape TM.
As the 2 tape T.M was already equiv is pow to an ordinary T.M
Sim of 2x ∞ tape T.M D w/ 2 tape TM M_1'
1st tape of M_1 is written w/ input str & the 2nd tape = blank

- Cut the tape of 2x ∞ tape T.M D into parts, at the starting cell of input str.

1 pt has input str & all blank space = right
2nd pt containing left of the input str appears on the 2nd tape in reverse order.

M sim D.
Thus 1 & 2

D is equiv to M. T.M w/ 2x ∞ tape is equiv to an ordinary T.M

TM Closure: 3.15c, 3.15d, 3.15e

3.15 Show that the collection of decidable languages is closed under the operation of

[redacted]

[redacted]n.

c. star.

d. complementation.

e. intersection.

3.15c) Let L be a Turing decidable Lang. & M be the T.M that decides L.

There's a T.M M' such that, it decides of L i.e, $L(M') = L^*$

Desc of M' is:

M' = on input w:

1. Split w into n pts such that $w = w_1 w_2 \dots w_n$ in different ways.
2. Run M on w_i for $i = 1, 2, \dots, n$.
   If M accepts each of these str $w_i \rightarrow$ accept
3. All cuts have been tried w/out success then reject.

When w is cut into differ substrings such that every str. is accepted by M, then w belongs to the star of L & thus M' accepts w after finite # of steps, else w will $\rightarrow$ reject, b/c there are finitely many possible cuts of w, M' will halt after finitely many steps.

Therefore, $L(M') = L^*$. The decidable Lang. are closed under star.

3.15d For a turing decidable Lang. L. T.M decides Lang M then the complement is M' on input w. The desc of M' is as follows

M' = on input w:

1. Acceps if M rejects
2. Else accept

Since M' Does the opposite of whatever M does, it decides the complement of L.
Therefore, decidable Lang. are closed under complementation.

3.15e Let L_1 & L_2 be 2 turing decidable Lang
M_1 & M_2 be the T.M that decides L_1 & L_2 respectively
There is a T.M M' such that, it decides intersection
Of L_1 & L_2 i.e $L(M') = L_1 \cap L_2$
Desc of M'

M' = on inupnt w;
1. Run M_1 on w. If M_1 rejects then reject.
2. Else run M_2 on w. If M_2 rejects then reject
3. Else accept.

M' Accepts w if both M_1 & M_2 accepts, if, if either Of the rejects then $M' \rightarrow$ rejects w,
Thus, $L(M') = L_1 \cap L_2$. The decidable lang are closed under intersect.